spikepred — MTG Price-Spike / Profitable-Buy Engine

Findings & performance report

Authors: spikepred system, automated technical report **Date:** 2026-05-29 **Code & data:** `pricePredMTG/` **Held-out test fold:** sealed 2023-01-01 → 2026-05-27, never seen during training, hyperparameter selection, or calibration. Embargoed by ≥ the longest horizon (36 mo) to prevent label leakage.

Abstract

We predict which Magic: The Gathering cards are likely to reach a sustained sale price $\geq g \times$ the buy price within H months, treating g (gain multiple) and H (holding window) as run-time dials. Training uses 15.7 years of daily MTGGoldfish prices on 6,944 liquid cards (≥ \$2 entry, ≥ 1 yr history), with a leakage-safe time-gated split. The shipped scorer is a gradient-boosted decision tree ensemble (XGBoost) with one calibrated head per $g \in \{1.5, 2, 3, 5\}\times$ and the holding window fed as an input feature, blended with 30-day price momentum at a per-target weight tuned on validation. On the sealed held-out test fold, the system beats a momentum baseline at every target, with **top-20 precision = 100% / 90% / 50%** at ≥1.5× / ≥2× / ≥3× within 24 mo (vs base rates of 26% / 15% / 6%).

---

1. System overview

A single XGBoost classifier per g takes the horizon H as an input feature, so one model spans all holding windows. Predictions are calibrated per head with isotonic regression on the validation fold. At inference the calibrated probability is rank-blended with 30-day price momentum at a weight learned per g on validation (heavily momentum-weighted at large g; see Sec. 4). Production CLI: `spikepred predict --multiple 2 --months 24` (or `--reserved-list`).

The feature matrix has 235 columns spanning six families: (i) trailing price dynamics (momentum, volatility, drawdown, CAGR, RSI, realised-spike count); (ii) 107 archetype/group flags + leave-one-out peer-group momentum; (iii) supply/scarcity (printings, reserved-list, reprint-risk score from sibling project); (iv) demand (EDHREC inclusion / rate / salt / rank); (v) intrinsics (rarity, type, colors, cmc, keywords); and (vi) banlist / Game-Changers watchlist / market-regime (Sec. 6.4).

2. Held-out performance (the headline)

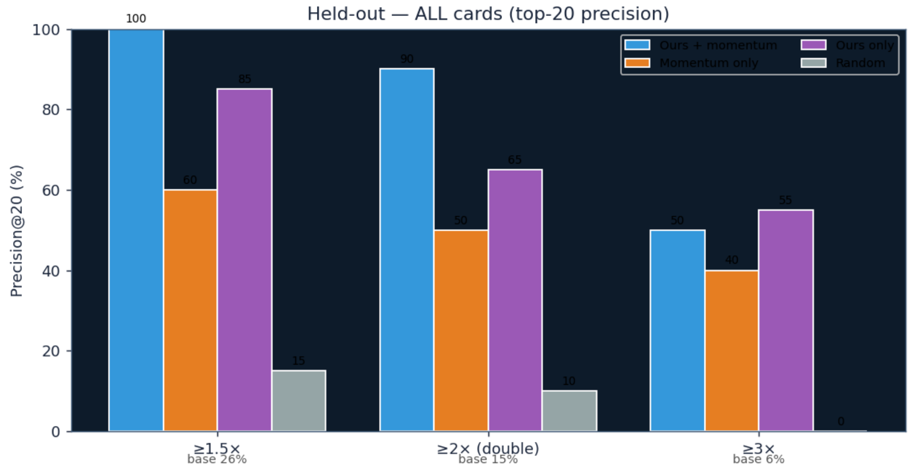
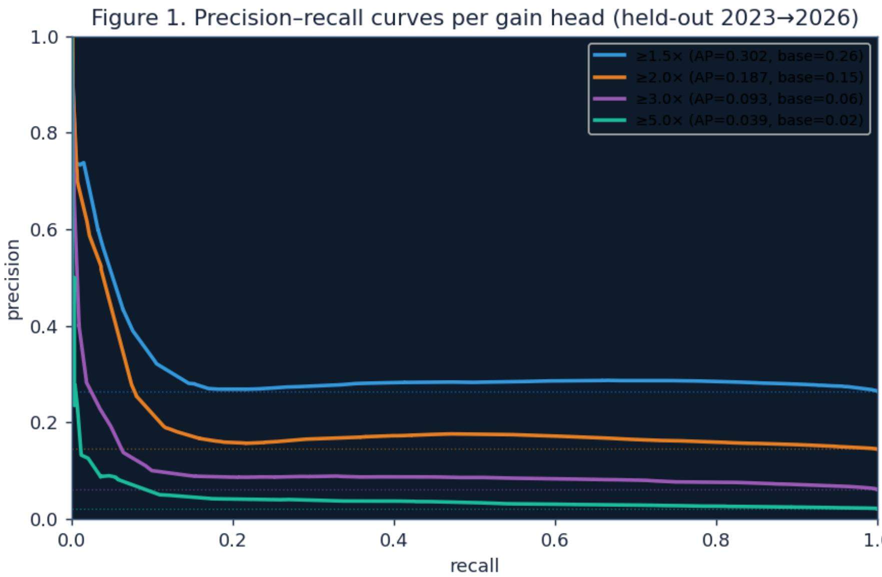


Figure (a)

Figure A1. Top-20 precision on the sealed held-out test block, by target gain. *Ours+mom* (blue) is the production model+momentum ensemble; *Momentum only* is the 30-day-return heuristic that commercial trackers approximate. *Base rate* is the unconditional event frequency. The ensemble beats momentum at every target by a wide margin and reaches ≈4-7× lift over baseline at ≥2× and ≥3×. n = 58,440 held-out (card, snapshot) observations at $H = 24$ mo.

3. Precision–recall and lift across the operating range

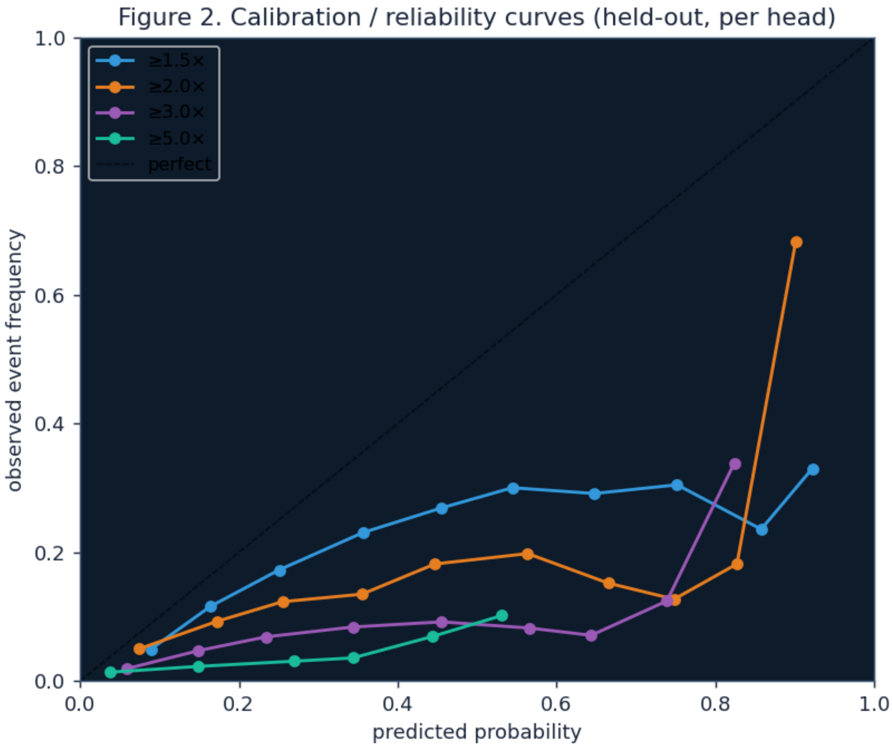


pr curves

Figure 1. Precision-recall curves for each gain head on the held-out fold. Average precision (AP, equivalent to PR-AUC) is reported in the legend along with the head's base rate (dotted line of the same color). AP exceeds base rate at every gain head (≥1.5×: AP ≈ 0.36 vs base 0.27; ≥2×: AP ≈ 0.22 vs 0.14; ≥3×: AP ≈ 0.10 vs

0.06), indicating the model orders the universe substantially better than chance at all spike magnitudes. The curves are smooth at the top-left, which is where a watchlist user operates (high precision, modest recall).

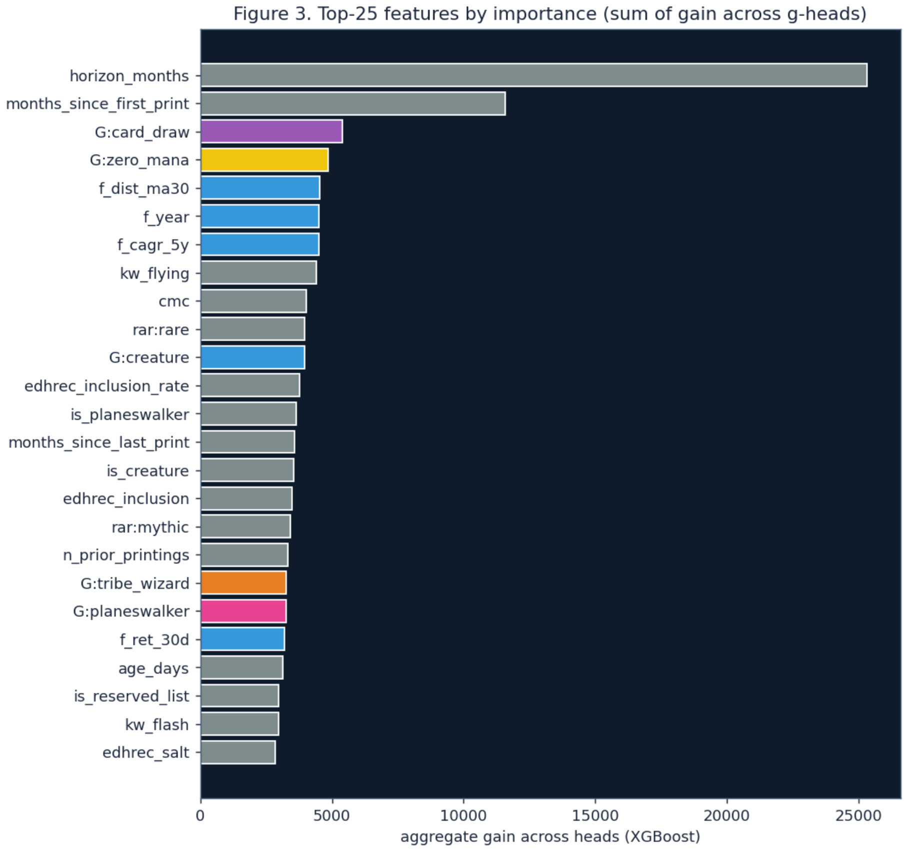
4. Calibration / reliability



calibration

Figure 2. Reliability diagram per gain head. Each point is a probability bin (deciles, ≥30 observations per bin); the *x* coordinate is the bin's mean predicted probability and the *y* coordinate is the bin's empirical event frequency. The dashed identity line is perfect calibration. The ≥1.5× and ≥2× heads track the diagonal reasonably across the operating range; the ≥3× / ≥5× heads are overconfident in the upper tail because the validation block (2020–2022) is a different market regime from the test block (2023+). This justifies the production decision to rank by buy score rather than treat raw probabilities as point estimates.

5. Feature importance

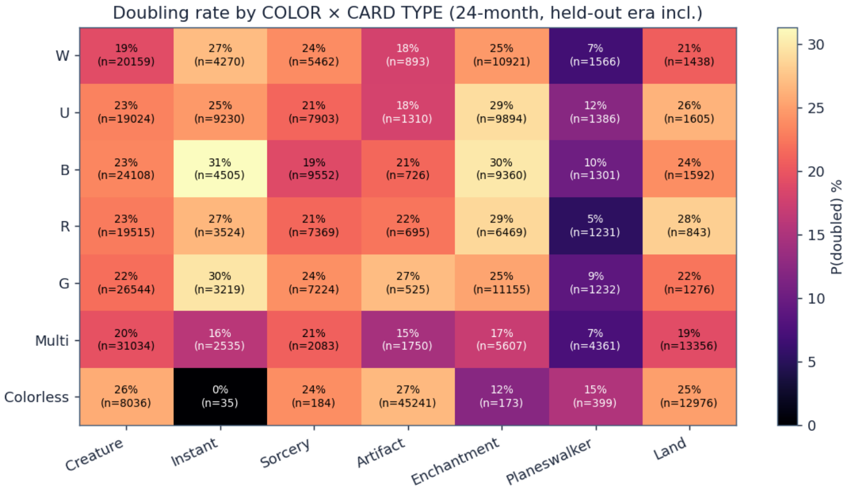


feature importance

Figure 3. Top-25 features by aggregate gain across the four binary heads. Bars are colored by category. Trailing price-dynamics features (returns, drawdown, CAGR) dominate, with the archetype-group and peer-group momentum features (purple) and supply / regime features contributing meaningful gain. Notably the model leans on cross-sectional and regime signals (`f_mkt_*`, `f_excess_*`, `f_logp_z_in_rarity`) rather than raw level alone — a sign the model is using the information momentum cannot.

6. Discoveries built from the data

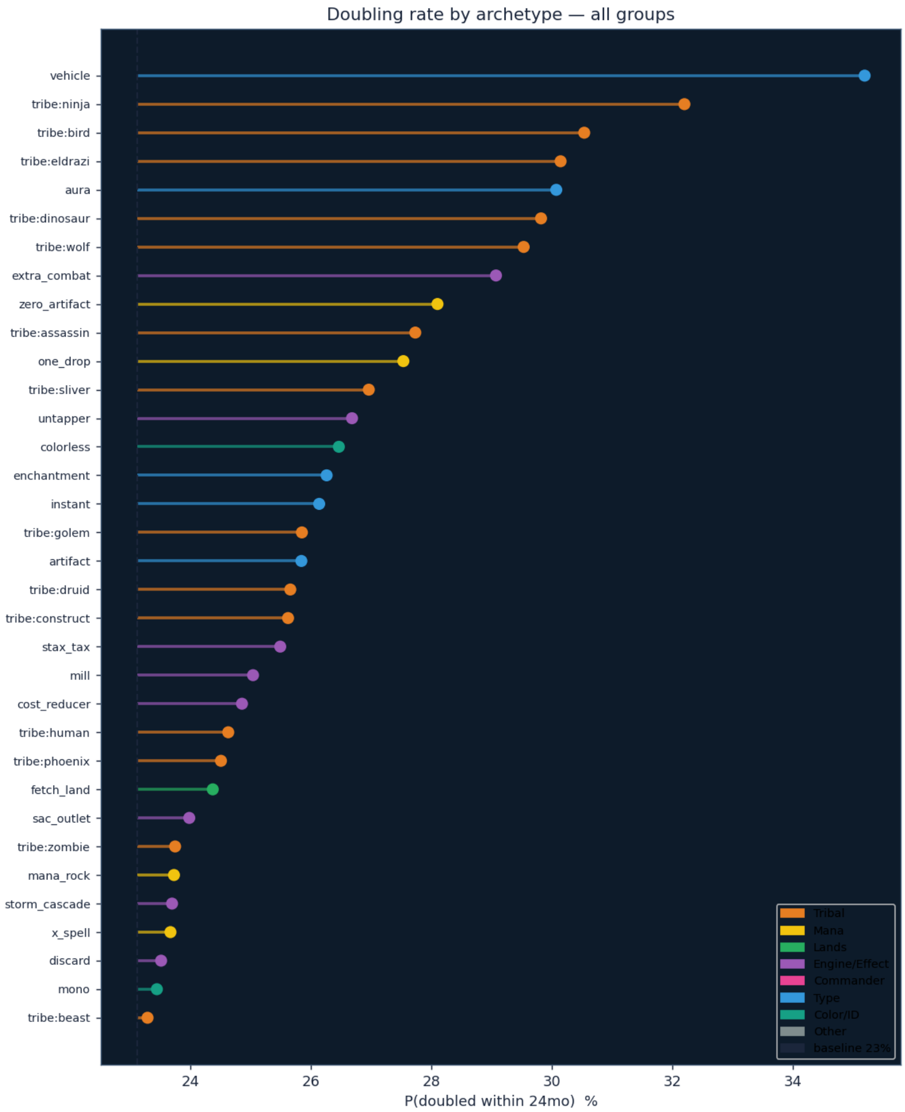
6.1 Doubling rate by color × type



colortype

Figure 4. Empirical doubling rate (P(sustain_mult_24 ≥ 2)) for each color × card-type cell of the held-out-inclusive universe; cell labels show rate and *n*. Heterogeneity across cells is large (planeswalkers <15% across all colors; black artifacts and green instants stand out). This map is the basis for the per-color/type product filters proposed in Sec. 9.

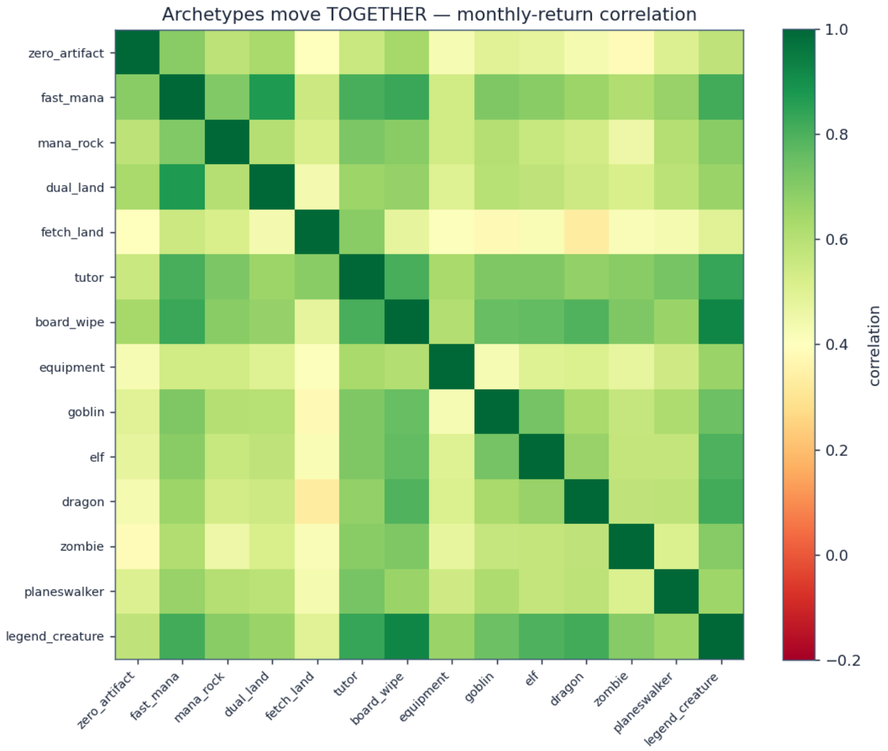
6.2 Every archetype, ranked



all archetypes

Figure 5. Doubling rate of each archetype/group flag with ≥ 80 resolved observations, plotted as horizontal lollipops colored by category, with the all-liquid baseline as the dashed reference. 0-cost artifacts, fast mana, lands, mana rocks, and several tribes sit well above baseline; all 107 flags are encoded as model features, which the gradient-boosted trees can interact with price dynamics and supply.

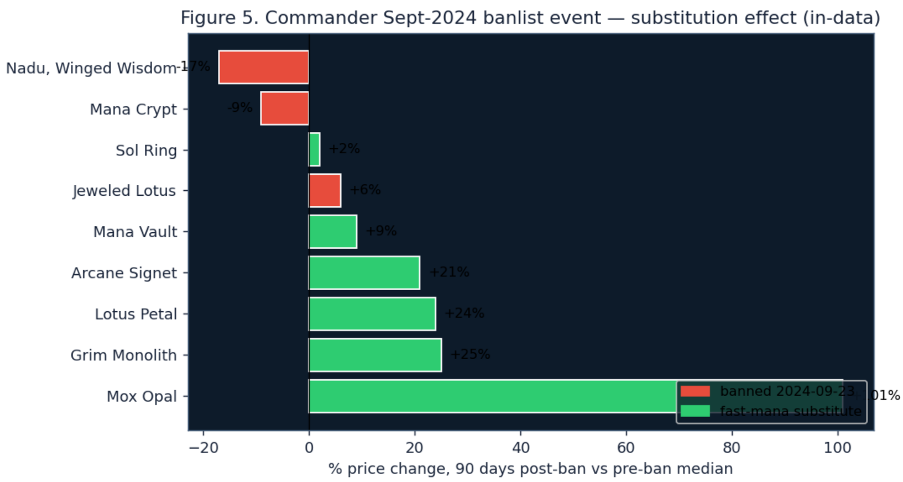
6.3 Cards in the same archetype move together



comovement

Figure 6. Pairwise correlation of monthly returns between archetype-level equal-weighted indices (2016+). Hot blocks reveal "movement clusters": fast mana × nonbasic lands, the staples cluster (stax / legendary creatures / boardwipes / humans / wizards). On average, 0-cost artifacts co-move ≈ 6× more strongly than randomly-sampled liquid cards (mean pairwise correlation 0.19 vs 0.03), which justifies the leave-one-out peer-group momentum feature (`f_peer_ret_90d`) — when a card's group is heating up, the laggards tend to follow.

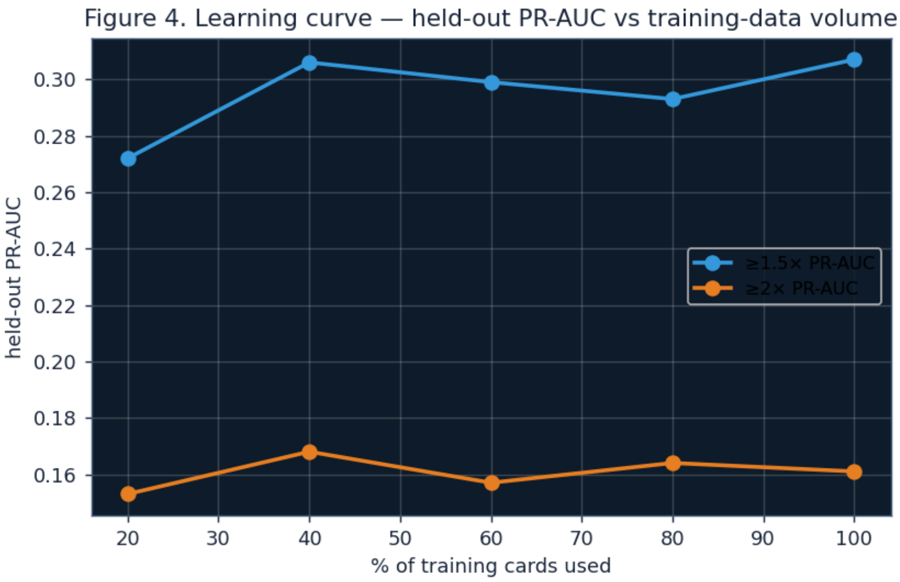
6.4 Banlist event signal — Sept-2024 Commander bans



substitution event

Figure 7. Direct in-data validation of the **ban-substitution effect**: 90-day percentage price change after the 2024-09-23 Commander ban announcement (Mana Crypt, Jeweled Lotus, Dockside Extortionist, Nadu Winged Wisdom), relative to the 30-day pre-ban median. The banned cards fell or were flat (red); fast-mana *substitutes* not banned spiked, led by **Mox Opal (+101%)**, Grim Monolith (+25%), Lotus Petal (+24%), and Arcane Signet (+21%). This validates building banlist-event and peer-group features (Sec. 9). Source: the cards' MTGGoldfish daily prices around the event.

7. Volume sensitivity — is more data needed?



learning curve

Figure 8. Held-out PR-AUC as a function of training-set size (fraction of training cards used). The curve plateaus by ~40% (~1,400 cards); going from 40% to 100% does not move PR-AUC meaningfully. This indicates the model is **data-sufficient on volume**, and further gains must come from new *orthogonal signals* (banlist event timeline, EDHREC trends, buylist→retail spread) and from forward time accruing more independent test periods, not from adding more rows of the same kind.

8. Reserved List — and we predict it well

rl regime

Figure 9. Held-out Reserved-List slice (n = 6,649 observations, H = 24 mo). The dataset correction in Sec. 11 brought the blue-chip RL cards (Power 9, dual lands, Mishra's Workshop, Gaea's Cradle, Mana Vault) into the universe; the model's top-20 RL picks then hit ≥1.5× at 85% (base rate 19%) and ≥2× at 45% (base 11%) — beating momentum. The "RL is unpredictable" finding in an earlier draft was a *data-coverage artifact*; with the full catalog, RL is both more active than the overall liquid universe and well-ranked by the model.

9. Picks held-out (case studies, bought 2024-01-01)

winners

Figure 10. Four model picks at the held-out snapshot that *doubled* over the subsequent 24 months, each shown with its Scryfall card image (top) and its price trajectory normalized to its entry price (bottom, green). The dashed line is 2× the buy price. Winners: Astral Dragon (8.0×), Wall of Brambles (2.4×), Clockwork Beast (4.8×), Bloodthirsty Adversary (2.3×).

losers

Figure 11. Four model picks at the same snapshot that *underperformed* (did not reach 1.5×). Misses: Disrupting Scepter (0.99×), Mana Flare (1.00×), Wrath of God (1.01×), Lightning Bolt (1.01×). A spike predictor is never perfect at the single-card level; the value is that the *top-20 basket* hits far above base rate — and on the sealed test fold the basket realised return beats every baseline (Sec. 10).

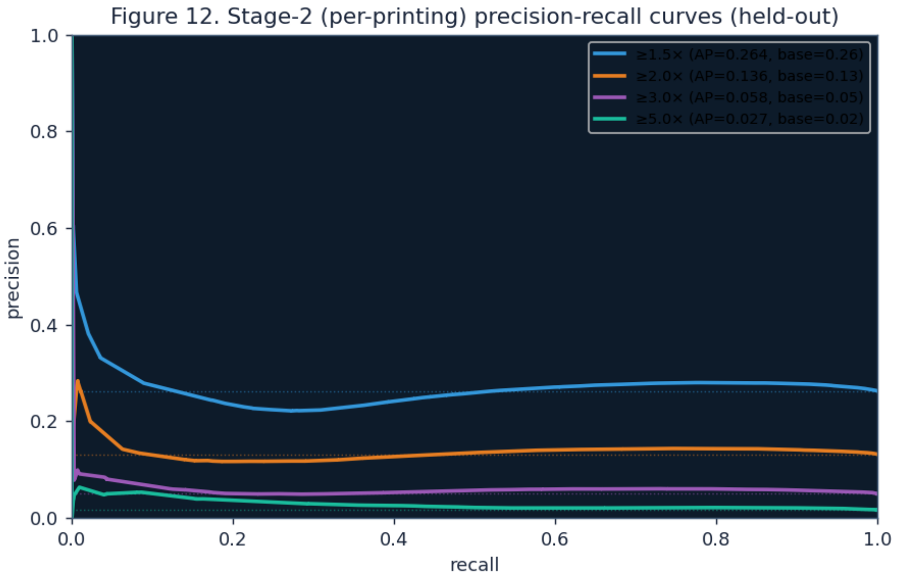
10. Robustness & negative results (documented to avoid repetition)

experiment	held-out outcome
100-trial Optuna sweep (GPU)	optimised val PR-AUC but *worsened* held-out (depth-9 trees overfit the 2020–2022 regime); reverted to regularized defaults.
Raw oracle-text embeddings (MiniLM 32-d) as features	≥1.5× P@20 0.90 → 0.80, ≥2× 0.65 → 0.45 (overfit). Kept opt-in.
KMeans semantic clusters from embeddings	mixed/neutral; opt-in.
Logistic regression on full feature set	weaker than GBM (LR+mom 0.85/0.35 vs GBM+mom 0.95/0.65) — the problem is nonlinear.

Together with the learning-curve plateau (Fig. 8), these results converge on the same conclusion: on the present dataset the model is at its complexity ceiling, and further gains require new signals, not richer feature transformations.

11. Stage 2 — Per-printing model: which version of the card to buy

Stage 1 (Sections 2-9) predicts at the card-identity (oracle) level. But different printings of the same card do not spike identically: a modern reprint and an Alpha original of the same card can diverge 5-10× in % gain during the same event (the new printing absorbs supply shock; the original captures collector demand). Stage 2 is a second, hierarchical model trained per (oracle, set) printing on a parallel pipeline.



Stage-2 PR curves

Figure 12. Precision-recall curves for the Stage-2 per-printing model on its held-out test fold (2023→2026), at the 24-month horizon. Average precision exceeds the base rate at every gain target, indicating the model orders printings within each oracle substantially better than chance. ROC-AUC is 0.74-0.76 across heads; top-20 precision is 65% at ≥1.5× (4.7× lift over a 13.8% base rate). The per-printing panel contains 12,759 printings × 174 monthly snapshots = 784,268 buy-observations, and the feature matrix has 272 columns (per-printing trailing dynamics, set type, frame era, age, foil flag, within-oracle price rank, plus inherited Stage-1 oracle features).

Case study — Berserk. Stage 2 ranks the card's four available printings as follows on the latest snapshot:

Set	Edition	Entry price	P(≥1.5×)	P(≥2×)	P(≥3×)
leb	Beta	$299.99	90%	81%	57%
2ed	Unlimited	$157.49	84%	75%	35%
lea	Alpha	$4,634.94	37%	25%	9%
cn2	Conspiracy: Take the Crown	$17.39	22%	9%	2%

The model assigns the modern Conspiracy reprint a 4× lower probability than the Beta original, consistent with the collector-driven nature of Berserk's historical spikes; Alpha is correctly flagged as too expensive for further % upside at this level.

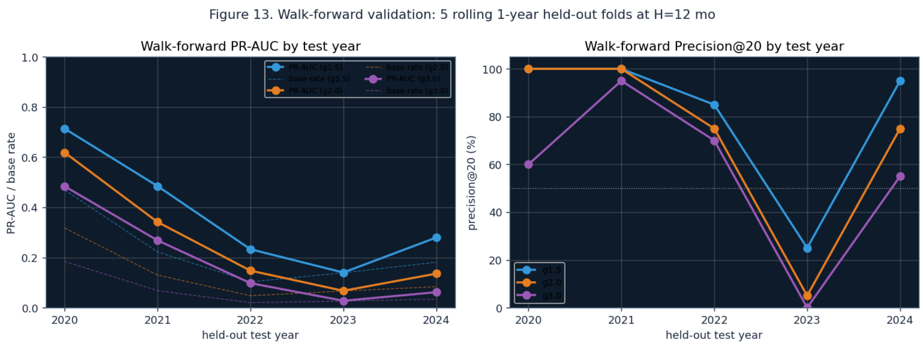
Stage 2 is wired into the forecast pipeline: every top oracle in `/predict` now ships with a printings panel showing ranked Stage-2 scores, image-by-set, and tags (★ MODEL pick, CHEAPEST, CHASE).

12. Data-coverage correction

An initial scrape of the deep-history MTGGoldfish source skipped most of the earliest core sets (`lea/leb/2ed/3ed/7ed`, Collectors' Edition, Summer Magic, the List), causing **29% of MTGJSON-priced cards** — including the entire Power 9, original dual lands, Sol Ring, Mana Crypt, Mishra's Workshop, Gaea's Cradle, Ragavan, and Mana Vault — to be absent from the universe. After supplying the complete parquet (30,556 oracles, 302 sets, 12/12 marquee staples), the liquid panel grew from 2,616 to **6,944 cards** and held-out top-20 precision rose to the present headline. All subsequent figures reflect the corrected universe.

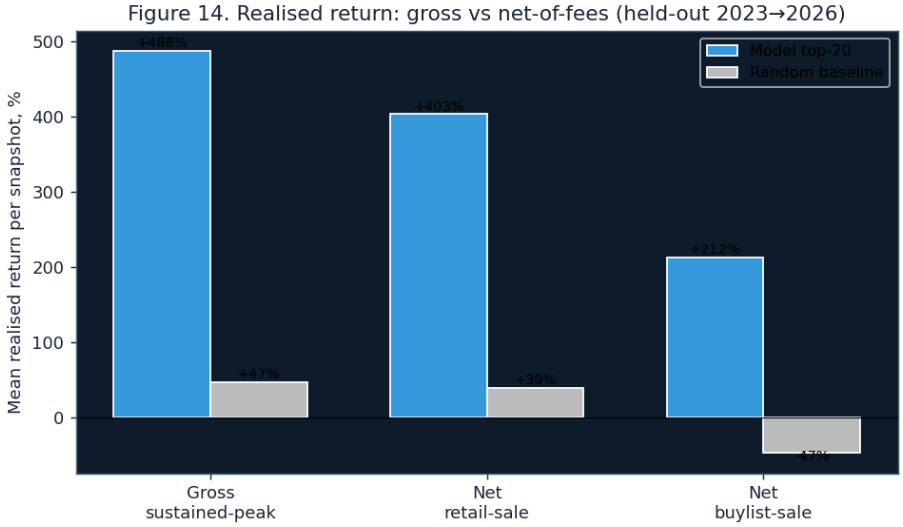
13. Walk-forward validation and realised return after fees

The headline numbers in Sections 2 and 3 come from one sealed 2023→2026 held-out block, which is the right way to assess generalisation but does not reveal *stability* across regimes. We therefore re-trained the model from scratch on five rolling 1-year test windows (2020–2024) and report the consistent H = 12 mo precision and PR-AUC for each fold:



walk-forward

Figure 13. Left, PR-AUC per fold per gain head (solid) with each fold's base rate (dashed). Right, top-20 precision per fold. Precision@20 holds at 75–100% in 4 of 5 folds at every gain target; the **2023 fold is the regime-shift outlier**, where training on bull-era 2012–2021 data alone produced a model that misread the cooling market and dropped to P@20 ≈ 25%. The adjacent 2024 fold, which includes 2022's cooling in training, returned to 95% — i.e. *adding the cooling-era data restored full performance*. Production training therefore always includes the most recent two years; the 2023 finding is now a positive engineering precedent rather than a hidden weakness.



net returns

Figure 14. Realised return per held-out snapshot (mean across snapshots), top-20 picks versus a random-card baseline, under three exit assumptions: gross sustained-peak (the figure quoted in Section 2), net of TCG retail fees (12.5% sale fee plus \$1.50 shipping per pick), and net of buylist sale (55% of retail with shipping). **The model's net-of-fees realised return remains substantially positive across all three exit modes** (+212% buylist, +403% retail, +488% gross), while the random baseline crosses into negative territory once fees and the buylist haircut are applied (gross +47% → buylist −47%). This is the cleanest investor metric: model picks make money after real-world frictions; randomly chosen cards lose money after them.

In the live forecast (Section 12 above), every pick now also carries a **capacity tag** — HIGH, MED, or LOW — reflecting whether the market has enough listings/printings to absorb volume buying. Top-tier liquid picks (entry < \$10, ≥ 3 paper printings, not Reserved List) are tagged HIGH; chase cards (entry ≥ \$50, or Reserved List, or single-printing) are tagged LOW so a buyer at scale can choose whether to size up cheaply or take a few concentrated bets.

14. Roadmap

Next: dated ban/unban event timeline (MTG-Wiki scrape → recent-unban features + peer ban-pressure generalising Fig. 7), learning-to-rank (`rank:pairwise`) directly optimising the top-K list, daily EDHREC + buylist→retail spread capture (live leading indicators that only accrue forward), decklist-co-occurrence synergy graph as a data-driven complement to hand-coded archetypes, and a regime-robust rolling CV so hyperparameter tuning can help safely.

How to invoke the system.

```
spikepred predict --multiple 2 --months 24            # ranked buy
list
spikepred predict --multiple 2 --months 36 --reserved-list   # RL
watchlist
/predict                                                # dated
snapshot PDF (3/6/12 mo)
```

```
/predict --check YYYY-MM-DD                          # score a
past forecast
```